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Zero-Shot NER in Slavic Languages: The Impact of Linguistic Relations, Lemmatization, and Transliteration

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Dimitar Kochev Mariia Zviahintseva Mihael Stoyanov Tetiana Tretiak
IT University of Copenhagen, Denmark
diko@itu.dk, mazv@itu.dk, svst@itu.dk, tetr@itu.dk

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Abstract

In this project, we explore how to improve cross-lingual named entity recognition (NER) for Slavic languages. We focus on performing zero-shot learning experiments that rely on the linguistic relations between four languages - Russian, Bulgarian, Ukrainian, and Slovene. To address challenges related to alphabet differences and rich morphology, we apply two preprocessing techniques: transliteration of Slovene from Latin to Cyrillic script and lemmatization across all languages in combination with the XLM-R model configuration. Our models' results support our initial hypothesis with a best F1 score of 89% for a Bulgarian model predicting on Slovene and 82% for a Russian model predicting on Ukrainian.

Our project repository: [GitHub](#)

1 Introduction

Named Entity Recognition (NER) is one of the fundamental tasks in natural language processing that involves identifying and classifying named entities such as people, organizations, and locations in the text. While current NER models have achieved significant accuracy in high-resource languages (e.g., English or Russian), performance often decreases significantly for low-resourced languages, such as many in the Slavic language group, e.g., Slovene and Ukrainian.

To address this problem, researchers started implementing the so-called zero-shot learning method, which involves training a model on one or more high-resourced languages and testing directly on data from a low-resource target language. This can help reduce the cost and effort of data annotation.

Languages from the Slavic family are an interesting case for this. They are linguistically related and have some fundamental similarities in grammar and vocabulary. However, they have some significant differences, such as the scripts they use. Ukrainian, Russian, and Bulgarian use the Cyrillic alphabet, while Slovene uses the Latin script. In this case, transliteration can improve model alignment across languages.

Another challenge is the morphological complexity. Many Slavic languages are highly inflected, meaning a single word can appear in many forms depending on case, gender, and number, which can confuse NER models. To overcome this, lemmatization, a process that reduces words to their dictionary (base) forms, can be applied.

In this project, we aim to improve the cross-lingual NER performance for Slavic languages by exploring the combined effects of transliteration, lemmatization, and linguistic similarity. We develop four models based on the XML-R-large configuration in HuggingFace: one trained only on Bulgarian data (bg-ner), one only on Russian data (ru-ner), one with Bulgarian and Russian (bg+ru-ner), and one on all four languages. The analysis of our results supported some of our linguistic assumptions, and it suggests some strategies for solving the NER task across Slavic languages.

2 Dataset

For this project, we use the dataset (3rd Shared Task on SlavNER) provided by the BSNLP 2021 Shared Task organizers to train and evaluate our models. The data consists of news articles on different topics collected from web sources. The annotation files follow the Universal NER convention. The dataset includes text in six Slavic languages: Ukrainian (uk), Slovene (sl), Russian (ru), Polish (pl), Czech (cs), and Bulgarian (bg). Five named entity categories are used in the annotation: PER (person), LOC (location), ORG (organization), EVT (event), and PRO (product). We focus on four languages:

Language	Train	Validation	Test
Russian (ru)	7,381	5,045	10,486
Bulgarian (bg)	11,715	3,104	3,384
Ukrainian (uk)	2,211	1,850	1,493
Slovene (sl)	7,512	5,186	4,312

Table 1: Sentence counts per language and dataset split

3 Related Work

Among the subtasks of the BSNLP 2021 workshop were both the identification and lemmatization of named entities. This inspired us to apply lemmatization to the training data for cross-lingual prediction. As discussed in (Pałka and Nowakowski, 2023), lemmatization can be particularly important for languages with rich inflectional morphology. For instance, in Russian, a name such as Иван (Ivan) can appear with various suffixes (-a, -e, -om, -y), depending on grammatical case. In contrast, the same name remains unchanged in Bulgarian. Consequently, if a model is trained on a language like Bulgarian and tested on Russian, the absence of lemmatization may lead to

misclassifications or missed entities due to inflectional variation. To mitigate this, we applied the Stanza lemmatizer uniformly across all data before training our models.

As for the choice of the model configuration, previous work on similar tasks (see references (Ulčar et al., 2021), (Viksna et al., 2023), (Arkhipov et al., 2019) and (Pałka and Nowakowski, 2023)) has evaluated multiple options. We opted for XLM-R large (Conneau et al., 2020), as it consistently outperforms or matches the performance of Multilingual BERT in cross-lingual tasks, making it a more suitable choice for our NER setup.

4 Linguistic Background

4.1 Linguistic Similarity

The language classification provided by Glottolog (Glottolog, 2024) offers insight into the relationships among various languages. The Slavic language family falls within the broader Indo-European family, specifically under the Classical Indo-European branch. It is further divided into three subgroups: East, West, and South Slavic. Our research focuses exclusively on the South and East Slavic languages. Russian and Ukrainian belong to the East Slavic subgroup, whereas Bulgarian and Slovene are part of the South Slavic subgroup. This classification suggests that Russian and Ukrainian are more closely related to each other than to Bulgarian or Slovene, and vice versa. For humans, this linguistic proximity often corresponds to higher mutual intelligibility between more closely related languages. Based on this, we hypothesize that such language similarity can be exploited in cross-lingual NER tagging. Specifically, we expect that a model trained on Bulgarian will perform better on Slovene than a model trained on Russian.

4.2 Writing Systems

As previously mentioned, one of the languages we focus on in our research uses a Latin-based script, while the other three use Cyrillic. To explore whether script alignment could enhance Slovene’s prediction accuracy, we investigated transliteration’s impact by converting Slovene data into Cyrillic. We hypothesize that transliterated Slovene will yield better results due to the alignment with the Cyrillic script used by the training languages.

To be more specific, since Bulgarian is linguistically the closest to Slovene, we chose to use a Bulgarian-based version of the Cyrillic alphabet. The transliteration was implemented using a custom-built tool. Most Latin letters have direct Cyrillic equivalents (e.g., b → б), but the letter j posed a challenge. Its representation varies by context: it can correspond to й or ъ, or combine with certain vowels to form new characters. For instance, when j follows l or n, it softens the consonant (e.g., lj → лъ, nj → нь). When paired with vowels like a or u, it becomes я (ja) or ю (ju), respectively. In all other cases, it is transliterated as й.

5 Experiment Details/Methodology

Some of the settings in our experiments and some of the code we developed were inspired by the following notebook (Paul, 2023).

5.1 Preprocessing

To perform the transformations mentioned in our previous sections, we constructed the following pipeline that has all of the steps required to prepare our data for training:

- **Lemmatization** – applied to each language file.
- **Transliteration** – we apply this only to the lemmatized Slovene data.
- **Tokenization** – our tokenizer is the pre-trained tokenizer from XLM-R-large. It is said to contain 250,000 tokens generated after analyzing multilingual data. This is perfect for our settings because it can cover both Cyrillic and Latin scripts.

The final output we pass to the model is a collection of tokenized sentences, an attention mask, and the respective BIO label of each token.

5.2 Models

To build our NER tagging system, we use the body of the XLM-R configuration and attach a customized head that can perform token classification. We also use the default AdamW optimizer, since it maintains steady performance when handling large models like the Transformers.

5.2.1 Model selection

In our Linguistic analysis section, we talked about the relation between the major Slavic groups (East and South Slavic). To evaluate the significance of their linguistic proximity, we decided to select one representative of each group (Bulgarian and Russian), fine-tune a model, and perform zero-shot testing. For example, we expected that our Bulgarian model would perform the better compared to our Russian model when testing on Slovene data. The same strategy was used when evaluating on Ukrainian data with the Russian model scoring higher.

After that, we decided to experiment with a combination of our training data. We created one model trained on a combination of our Bulgarian and Russian data. Finally, we created a more diverse model with all of our available data. The idea was to investigate how much the addition of a new language can increase the test performance above our previous benchmarks.

Once we built our four initial models, we decided that the significance of lemmatization should also be tested. This is why we added a model trained on Russian non-lemmatized data. We did not follow along with the Bulgarian, because it has little to no grammatical inflection.

As a performance metric, we decided to use the normal span F1 score.

5.2.2 Model hyper-parameters

- **Learning rate:** 3e-5
- **Batch size:** 16
- **Epochs:** 5
- **Weight decay:** 0.01
- **Warm-up ratio:** 0.1
- **Learning scheduler type:** linear

Learning rate – standard for the type of our task

Epochs – only the final model was trained on 3 epochs because of the restriction of our resources

Warm-up ratio – The use of warm-up steps is proven to lead to higher performance of Transformer models. This allows the learning rate to increase gradually and avoid poor convergence due to a rapid start.

5.3 Results

Evaluated on	ru	bg	uk	sl	sl_trans	Average
Fine-tune on						
ru	0.781	0.852	0.825	0.861	0.749	0.814
ru not lemmatized	0.744	0.834	0.796	0.843	0.681	0.780
bg	0.754	0.870	0.774	0.892	0.719	0.802
ru+bg	0.780	0.865	0.811	0.896	0.723	0.815
all	0.775	0.890	0.812	0.883	0.761	0.824

Table 2: Span F1 results of our models

After training the bg-ner, ru-ner, bg+ru-ner, and all-langs-ner models, we tested them on the available test sets for all four languages and the transliterated version of Slovene. We decided not to compare the performance of a single model across training sets for different languages, as variations in data quantity and quality significantly impact performance. Instead, we focus on identifying which model performs best for each specific language. The results support our hypothesis: the model trained on Bulgarian performed better on Slovene than the model trained on Russian. Similarly, for Ukrainian, the Russian model achieved the best zero-shot performance. These findings confirm that zero-shot NER performs better when the training and target languages are more closely related.

On the other hand, we concluded that transliterating Slovene decreased the models’ performance. This could be due to the fact that the transliterated words are not properly recognized by the XLM-R-large tokenizer, leading to incorrect tokenization and NER predictions. At the same time, all trained models achieved high accuracy on the original, non-transliterated Slovene data.

In addition, it proved to be true that training a model on a combination of languages can improve its robustness and, therefore, lead to better average performance across all four target languages. According to our results, the model trained on all languages achieved the highest average performance, while the ru+bg-ner model also outperformed both ru-ner and bg-ner.

We trained and tested the Russian model using both lemmatized and non-lemmatized data. The results showed that lemmatization improves performance, particularly for languages with rich inflectional morphology, by reducing variability and helping the model better recognize named entities.

Figure 1 shows that the zero-shot Russian model outperforms the model fine-tuned on a small Ukrainian dataset. At least 2,000 Ukrainian training samples are needed to reach a comparable F1 score, which can be challenging and time-

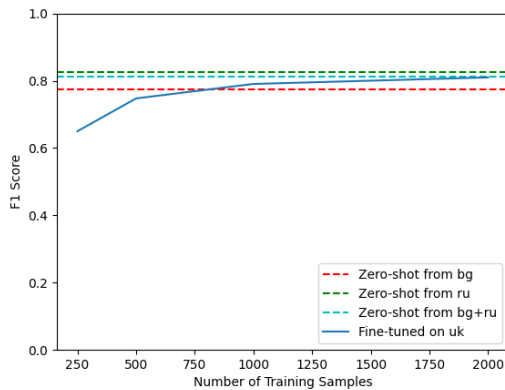


Figure 1: Ukrainian

consuming to obtain. Given that the performance improvement is not substantial, zero-shot learning can be considered a solid and practical solution for low-resource languages.

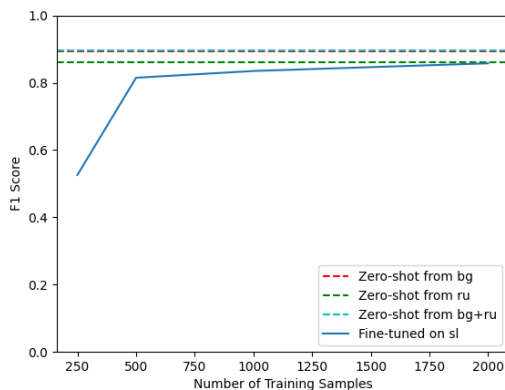


Figure 2: Slovenian

A similar trend is observed for Slovene, where the zero-shot model trained on Bulgarian already achieves high performance, and further fine-tuning on Slovene data provides only marginal gains.

6 Conclusion

In our project, we explored how linguistic relationships within the Slavic language family can enhance NER performance for low-resourced languages. To tackle this challenge, we utilized zero-shot learning, aiming to identify which high-resource languages yield the best accuracy on test sets from other Slavic languages and to determine which factors contribute to improved performance.

For each model, we built a processing pipeline that involved uploading training and validation datasets for each language, lemmatizing the data,

applying transliteration specifically for Slovene, tokenizing the text, and training the model using XLM-R-large.

Our experiments showed that lemmatization significantly improves model performance for Slavic languages, likely due to their rich inflectional morphology. In contrast, transliteration for Slovene did not lead to improvements in model performance.

We demonstrated that leveraging linguistic similarities within the Slavic language family can enhance zero-shot learning performance. As anticipated, incorporating multiple languages into the training data improves the model’s robustness and generalization capabilities, though it also increases annotation effort and training costs.

Finally, using a high-accuracy zero-shot NER tagger on the target language may be more effective and cost-efficient than fine-tuning a model on a low-resourced language.

6.1 Limitations

- Our dataset is derived from news articles across four topics. Although these articles use formal language, they are non-academic in nature. To obtain more broadly applicable results, we should expand the dataset to include a wider range of text types, such as informal and academic texts from various domains.
- For Slovene, transliteration could be beneficial for multilingual models, provided we have a pretrained tokenizer that can correctly tokenize transliterated words. Alternatively, with significantly more data, we could train a tokenizer ourselves. Yet this would make the solution useless for applications outside research, since a Cyrillic version of Slovene would never be used in the real world.
- We could further explore the language similarity hypothesis by expanding our dataset to include more South Slavic languages, such as Serbian and Croatian. These languages are even closer to Slovene, because they belong to the same Western South Slavic subgroup and they use the Latin script. Serbian also has an official Cyrillic transliteration, which makes it perfect for further investigation of script significance.

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